HDOC Day-72 Activity

Student Attendance Using a Dynamically Resizable Matrix

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Aim

To develop a C program that manages student attendance using a dynamically allocated and dynamically resizable matrix. The program uses pointers, malloc(), realloc(), and free(), and calculates the attendance percentage of each student.

Algorithm

1. Read the initial number of students and working days and validate that both are positive.
2. Create the initial attendance matrix dynamically using malloc().
3. Record attendance for each student and each day; accept only 0 (Absent) or 1 (Present).
4. Display the initial attendance matrix.
5. Read the number of additional students to be added.
6. Resize the row-pointer array using realloc(), allocate rows for the new students, and record their attendance.
7. Display the resized attendance matrix.
8. For each student, count the number of present days and calculate percentage = (present / working days) x 100.
9. Release every allocated row and then the matrix pointer using free().

C Program

```c
#include <stdio.h>
#include <stdlib.h>

int **createMatrix(int students, int days) {
    int **a = (int **)malloc(students * sizeof(int *));
    if (a == NULL) return NULL;

    for (int i = 0; i < students; i++) {
        a[i] = (int *)malloc(days * sizeof(int));
        if (a[i] == NULL) {
            for (int j = 0; j < i; j++) free(a[j]);
            free(a);
            return NULL;
        }
    }
    return a;
}

void recordAttendance(int **a, int start, int students, int days) {
    for (int i = start; i < students; i++) {
        printf("Enter attendance for Student %d:\n", i + 1);
        for (int j = 0; j < days; j++) {
            do {
                printf("Day %d (0/1): ", j + 1);
                if (scanf("%d", &a[i][j]) != 1) {
                    printf("Invalid input. Enter only 0 or 1.\n");
                    while (getchar() != '\n');
                    a[i][j] = -1;
                } else if (a[i][j] != 0 && a[i][j] != 1) {
                    printf("Invalid input. Enter only 0 or 1.\n");
                }
            } while (a[i][j] != 0 && a[i][j] != 1);
        }
    }
}

int **addStudents(int **a, int oldCount, int addCount, int days) {
    int newCount = oldCount + addCount;
    int **temp = (int **)realloc(a, newCount * sizeof(int *));
    if (temp == NULL) return NULL;
    a = temp;

    for (int i = oldCount; i < newCount; i++) {
        a[i] = (int *)malloc(days * sizeof(int));
        if (a[i] == NULL) {
            for (int j = oldCount; j < i; j++) free(a[j]);
            return NULL;
        }
    }
    return a;
}

void displayAttendance(int **a, int students, int days) {
    printf("\nStudent");
    for (int j = 0; j < days; j++) printf("\tDay %d", j + 1);
    printf("\n");

    for (int i = 0; i < students; i++) {
        printf("Student %d", i + 1);
        for (int j = 0; j < days; j++) printf("\t%d", a[i][j]);
        printf("\n");
    }
}

void calculatePercentage(int **a, int students, int days) {
    printf("\nAttendance Percentages:\n");
    for (int i = 0; i < students; i++) {
        int present = 0;
        for (int j = 0; j < days; j++) present += a[i][j];
        printf("Student-%d: %.1f%%\n", i + 1,
               (present * 100.0) / days);
    }
}

void freeMatrix(int **a, int students) {
    for (int i = 0; i < students; i++) free(a[i]);
    free(a);
}

int main(void) {
    int students, days, extra;

    printf("Enter initial number of students: ");
    if (scanf("%d", &students) != 1 || students <= 0) {
        printf("Invalid number of students.\n");
        return 1;
    }
```

```
    printf("Enter number of working days: ");
    if (scanf("%d", &days) != 1 || days <= 0) {
        printf("Invalid number of working days.\n");
        return 1;
    }

    int **attendance = createMatrix(students, days);
    if (attendance == NULL) {
        printf("Memory allocation failed.\n");
        return 1;
    }

    recordAttendance(attendance, 0, students, days);
    printf("\nInitial Attendance Matrix:\n");
    displayAttendance(attendance, students, days);

    printf("\nEnter number of additional students: ");
    if (scanf("%d", &extra) != 1 || extra < 0) {
        printf("Invalid number.\n");
        freeMatrix(attendance, students);
        return 1;
    }

    if (extra > 0) {
        int oldCount = students;
        int **temp = addStudents(attendance, oldCount, extra, days);
        if (temp == NULL) {
            printf("Memory allocation failed while adding students.\n");
            freeMatrix(attendance, oldCount);
            return 1;
        }
        attendance = temp;
        students += extra;
        recordAttendance(attendance, oldCount, students, days);
    }

    printf("\nFinal Attendance Matrix:\n");
    displayAttendance(attendance, students, days);
    calculatePercentage(attendance, students, days);

    freeMatrix(attendance, students);
    return 0;
}
```

Example Run

Initial attendance matrix (3 students, 3 working days)

Student	Day 1	Day 2	Day 3
Student 1	1	0	0
Student 2	1	0	1
Student 3	1	1	1

After dynamically adding 3 more students

Student	Day 1	Day 2	Day 3
Student 1	1	0	0
Student 2	1	0	1
Student 3	1	1	1
Student 4	1	0	0
Student 5	1	1	1
Student 6	1	0	1

Attendance Calculation

Student-1: 33.3% Student-2: 66.7% Student-3: 100.0%
Student-4: 33.3% Student-5: 100.0% Student-6: 66.7%

Result

The program successfully stores attendance in dynamically allocated memory, resizes the matrix when new students are added using realloc(), validates attendance input, calculates each student's attendance percentage, and releases allocated memory before termination.

HDOC Day-72 - Program Execution

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```
Last login: Tue Nov 10 22:44:57 on ttys000
dhruvpandey@DHRUVs-MacBook-Air ~ % nano day72.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang day72.c -oday72
dhruvpandey@DHRUVs-MacBook-Air ~ % ./day72
Enter initial number of students: 2
Enter number of working days: 1

Enter attendance for Student 1:
Day 1 (1 = Present, 0 = Absent): 2
Invalid input! Enter only 0 or 1.
Day 1 (1 = Present, 0 = Absent): 2
Invalid input! Enter only 0 or 1.
Day 1 (1 = Present, 0 = Absent): 3
Invalid input! Enter only 0 or 1.
Day 1 (1 = Present, 0 = Absent): 1

Enter attendance for Student 2:
Day 1 (1 = Present, 0 = Absent): 2
Invalid input! Enter only 0 or 1.
Day 1 (1 = Present, 0 = Absent):
```